Model Predictive Control for Adaptive Energy-Efficient Street Lighting: A Comprehensive Framework and Case Study

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***Abstract*—This study tackles the challenge of optimizing light emitting diode (LED) street lighting to enhance energy efficiency without compromising safety and comfort in dynamic urban environments. Conventional adaptive lighting systems often overlook real-time microlocation variations, such as fluctuating traffic, pedestrian activity, and weather conditions, resulting in excessive energy consumption or inadequate illumination. To address this limitation, we develop and validate a simulation environment that integrates a model predictive control (MPC) algorithm designed to predict and dynamically adjust lighting intensity using high-resolution environmental data. Our primary objectives are to establish a robust MPC framework for real-time streetlight optimization under varying conditions, to validate its performance through simulation and real-world deployment, and to quantify its energy savings. Theoretically, this work contributes to a novel MPC-based control strategy for predictive modeling of dynamic urban variables, allowing proactive lighting adjustments. Practically, we demonstrate the system's feasibility through a case study in Sisak, Croatia, where it achieved 31.94% energy savings while maintaining illumination quality. This research advances smart city infrastructure by offering a scalable and tunable control framework that harmonizes energy efficiency with public safety.**

***Index Terms*—Adaptive lighting systems, energy efficiency, LED street lighting, model predictive control, simulation environment, smart cities.**

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I. Introduction

URBAN lighting plays a vital role in infrastructure, with its energy consumption varying significantly across different regions. In some places, it could reach even up to 40% of municipal electricity costs. It is responsible for 15% of global power consumption, while contributing up to 5% to greenhouse gas emissions [1]. Conventional high-pressure sodium street lighting systems are often energy-intensive, costly, and environmentally harmful, while also being inflexible and inefficient in their design. Introduction of modern light emitting diode (LED) lights significantly improved street lighting efficiency but majority of installed lamps are unable to respond to dynamically changing conditions, resulting in unnecessary energy consumption and higher maintenance requirements.

A. Motivation

The lack of remote control and real-time adaptability of street lighting to dynamic traffic, weather, and time of day, exacerbates the system inefficiency. With energy demand expected to rise by 50% over the next two decades, innovative strategies are essential to reduce carbon footprints and operating costs. LED technology overcomes these challenges by reducing energy consumption by up to 40%, lowering emissions, and mitigating light pollution via precise directional lighting [2]. LEDs are efficient, durable, and adaptable (dimmable), which makes them ideal for modern public lighting and for supporting intelligent street lighting systems essential for smart cities. However, integrating advanced control mechanisms into these systems remains challenging. Building on previous work [3], presented at the 62nd IEEE Conference on Decision and Control (CDC), this study proposes a cost-effective urban street lighting solution combining energy-efficient LED technology with a low-cost embedded controller. Compared to the conference paper, this work includes new validation steps using real-world data from industry partner and scales implementation to a full street of 100 luminaires. Professional ray tracing software and experimental results validate the derived mathematical model for model predictive control (MPC). It is then utilized in annual energy simulations that refine the MPC algorithm, demonstrating improved energy efficiency and real-time adaptability.

Results from a year-long simulation highlight significant energy savings, optimized control actions, and enhanced lighting quality, validated through comprehensive analysis.

B. Literature Review

Recent advancements in street lighting technology have focused on developing intelligent, adaptive systems designed to improve energy efficiency and enhance urban safety. This literature review examines five key themes in current research: 1) intelligent adaptive systems; 2) multisensory and Internet-of-Things (IoT) integration; 3) MPC and advanced control strategies; 4) dynamic real-time adaptation; and 5) comprehensive system integration. These themes collectively inform the conceptual framework for our proposed MPC-based lighting control system.

1) Intelligent and Adaptive Lighting Systems: The evolution of intelligent street lighting is evident in systems such as those described in [4] and [5]. These systems utilize infrared (IR) and passive infrared (PIR) sensors to detect vehicles and pedestrians, adjusting lighting accordingly. For example, Al-Smadi et al. [4] presents a system that activates streetlights when vehicles are detected and deactivates them once they pass, conserving energy on low-traffic roads. Similarly, Dhamodaran et al. [5] employs microcontrollers to adjust lighting intensity based on movement detection, enhancing the adaptive response of the system. Incorporating pedestrian detection alongside vehicular traffic is a critical aspect of modern street lighting. The study in [6] demonstrates how integrating vehicle and pedestrian data into a street lighting network can produce energy savings proportional to traffic volumes. Complementing this, Xue et al. [7] introduces an intelligent dimming algorithm that uses a fuzzy neural network to optimize lighting based on variables such as vehicle speed and pedestrian density, further refining the system's adaptability.

2) Integration of Multisensory and IoT Technologies: Recent studies have explored the integration of multisensory inputs and IoT frameworks to enhance street lighting systems. For instance, Khalil et al. [8] describes a system that adjusts brightness based on combined data from traffic, pedestrian movements, and weather conditions. The inclusion of IoT management frameworks, as seen in [9], enables weather-adaptive lighting, which not only improves safety but also significantly reduces energy consumption. The application of advanced methodologies such as ray tracing and Monte Carlo methods in optimizing LED streetlight designs is well-documented. These techniques ensure compliance with photometric standards set by the commission internationale de lÉclairage (CIE) and illuminating engineering society (IES), focusing on refining LED lamp designs and placements to maximize efficiency and uniformity [10], [11]. Innovative streetlight configurations proposed in [12] and [13] have further enhanced the economic and environmental impact of lighting systems. The role of IoT in transforming street lamps into smart network nodes is pivotal in modern urban environments. Studies such as [14] and [15] explore how streetlamps can communicate via Wi-Fi to central systems, improving control efficiency. In addition, Agramelal et al. [16] details a predictive traffic-based smart lighting system that employs deep learning to anticipate and respond to changes in traffic flow, marking a significant step toward more responsive urban infrastructure. Cloud-based IoT platforms, as discussed in [17], allow for remote monitoring and control of street lighting systems, which is further enhanced by the integration of wireless sensor networks for real-time data collection and adjustments [18]. The studies in [19], [20], and [21] highlight the importance of integrating various sensors and communication technologies to optimize the functionality and efficiency of street lighting systems.

3) Advanced Adaptive Strategies: The incorporation of deep learning and fuzzy logic in controlling lighting intensity, as mentioned in [22] and [23], exemplifies the shift toward data-driven, intelligent management strategies. Efforts to enhance street lighting efficiency have increasingly focused on predictive and adaptive control mechanisms. These strategies, which respond dynamically to traffic and environmental conditions, not only reduce energy consumption but also improve safety and visibility [13].

4) Dynamic Control and Real-Time Adaptation: Recent advancements have leveraged modern technologies to improve the adaptability of street lighting systems. For example, Nair and Rahul [24] introduces a dynamic control system that uses a deep neural network-based camera to adjust luminance in real time, demonstrating significant improvements in light intensity management. Similarly, Ge et al. [25] developed an intelligent street lighting system that integrates data fusion technology to modify light intensity based on environmental conditions, with fuzzy logic enhancing system smoothness. Dynamic dimming strategies, as outlined in [26], utilize traffic flow predictions to optimize public lighting, achieving substantial energy savings. Optimistic scientific projections from these groups suggest that adaptive LED lighting—controlled by a central operator—could achieve 30–50% energy savings by adjusting to local conditions, particularly traffic density [2], [27]. Supporting evidence comes from artificial neural network traffic modeling and static lighting adjustments (e.g., 50% lighting reduction during nighttime), utilizing data from Terni, Italy [28] to Bari, Italy [29].

5) Comprehensive System Integration: Advanced intelligent power electronics management systems, such as those described in [30], [31], and [32], combine sensor-based identification, adaptive strategies, and dual control mechanisms to minimize energy use and enhance system performance. The integration of adaptive control systems into power electronics frameworks, as explored in [33], [34], and [35], underscores the potential for significant energy savings and improved responsiveness.

This comprehensive review reveals three critical insights that guide our study: First, while existing systems demonstrate the effective use of sensors and IoT for adaptive lighting, they typically operate with reactive rather than predictive control strategies. Second, MPC approaches show superior energy efficiency potential but have not been sufficiently explored for microlocation adaptation. Third, the integration of multiple data streams (traffic, pedestrian, and weather) within a predictive framework remains an underexplored opportunity. These

TABLE I
COMPARISON OF STREETLIGHT CONTROL METHODS

Reference	Method	Key Features	Limitations	Our Advancement
[4]	IR/PIR sensor-based	Vehicle detection, ON/OFF control	No pedestrian detection, reactive only	Predictive MPC with multi-objective optimization
[5]	Microcontroller + PIR	Movement-based dimming	Fixed dimming levels, no weather adaptation	Dynamic tuning of safety-energy tradeoffs
[6]	Traffic-volume adaptive	Savings proportional to traffic	No real-time optimization	MPC with 1–5 min resolution
[7]	Fuzzy neural network	Vehicle speed/pedestrians density	Heuristic rules, no MPC	MPC optimization
[8]	Multisensory IoT	Traffic/pedestrian/weather integration	Rule-based reactions	Predictive horizon optimization
[24]	Deep neural network	Camera-based real-time adjustment	High computational cost	Fast MPC (less than 1 s)
[25]	Fuzzy logic + data fusion	Environmental adaptation	No spatial optimization	Microlocation-aware control
[28]	Static dimming	50% nighttime reduction	Inflexible schedule	Real-time dynamic adaptation
[29]	ANN traffic modeling	Static traffic adjustments	No pedestrian safety	Integrated safety constraints

findings directly motivate our development of a microlocation-aware MPC system that synthesizes these capabilities into a unified control framework.

C. Article Contribution

This work introduces a novel MPC framework for microlocation-aware street lighting, integrating real-time traffic, pedestrian, and weather predictions, factors often ignored in conventional systems. Unlike rule-based methods, our optimization-driven approach dynamically adjusts lighting at fine spatiotemporal resolutions to balance energy efficiency, safety, and comfort. The main contributions are as follows.

1) *Theoretical Advancements*: Extends MPC theory to street lighting with a high-resolution, multiinput model for stochastic urban dynamics. Tunable parameters (e.g., prediction horizons and safety-energy tradeoffs) set a precedent for smart infrastructure.
2) *Practical Breakthroughs*: Real-world deployment in Sisak, Croatia, achieved 31.94% higher energy savings than current lighting system while meeting illumination standards. Low computational overhead and IoT compatibility enable scalable retrofitting.
3) *Field-Broadening Impact*: Bridges industrial MPC techniques with urban challenges of the public sector, advancing dynamic, data-driven lighting beyond static paradigms.

This work establishes a foundational toolkit for smart lighting, uniting energy efficiency and public safety through real-time, physics-informed optimization. An overview of the street lighting control methods, gap analysis, and our contribution is summarized in Table I.

The article is structured as follows. Section I reviews the literature, identifying gaps in predictive control and real-time adaptation. Section II outlines the streetlight illumination model and the MPC algorithm. Section III details the implementation with a case study from Sisak, Croatia. Section IV concludes with findings and future directions.

II. METHODOLOGY

A. Street Lighting Model

1) Street Configuration: A real-world test site was selected in collaboration with an industry partner, LED Elektronika, to demonstrate the practical implementation of the proposed MPC-based street lighting optimization system. The selected location, J. J. Strossmayer Street in Sisak, Croatia, serves as an exemplary urban setting. The street is equipped with dimmable LED luminaires (PrecisionLux3 LECS-120-WW-D3T, 120W), chosen for their energy efficiency and consistent performance. It features a two-lane configuration with single-sided luminaire distribution (type M4), meeting vehicular and pedestrian lighting standards. This setup provides an ideal environment for testing under varying conditions, including weather, traffic, and pedestrian densities. Integrating MPC control with these luminaires demonstrates a practical approach to energy-efficient, adaptive street lighting.

2) Mathematical Model of Illuminance: Under steady-state conditions, the illuminance at any point p in the three-dimensional space of the street is determined using a spatial coordinate system based on the geometric properties of each light source. This model, as defined in [36], relates the luminous intensity I as output of multiple luminaires to the illuminance E_p at target location with Cartesian coordinates (x, y, z). It incorporates factors such as luminaire height h and boom length d. The combined effect of n luminaires governs the overall illuminance at point p is given as

$$E_p(x,y,z) = \sum_{i=1}^{n} \frac{I_i(C_i,\gamma_i)\cdot h_i}{\left[(x^2)+(y-d_i)^2+(h_i-z_i)^2\right]^{3/2}} \tag{1}$$

where
E_p: Illuminance at the observed point in 3-D space, measured in lux (lx).
I_i: Angular luminous intensity of the i^{th} luminaire, measured in candelas (cd) with maximum values.
n: Number of light sources influencing the point.
C_i: Photometric azimuth of the light's path from luminaire i to point p.
γ_i: Vertical photometric angle of the light's path from luminaire i to point p.
h_i: Height at which the i^{th} luminaire is mounted.
d_i: Length of the boom on which the i^{th} luminaire is mounted.
Photometric angles are obtained by

$$C = 90^\circ + \arcsin\left(\frac{[(y-d)\cos\sigma - h\sin\sigma]^2 + x^2}{[(y-d)^2+x^2+h^2]^{1/2}}\right) \tag{2}$$

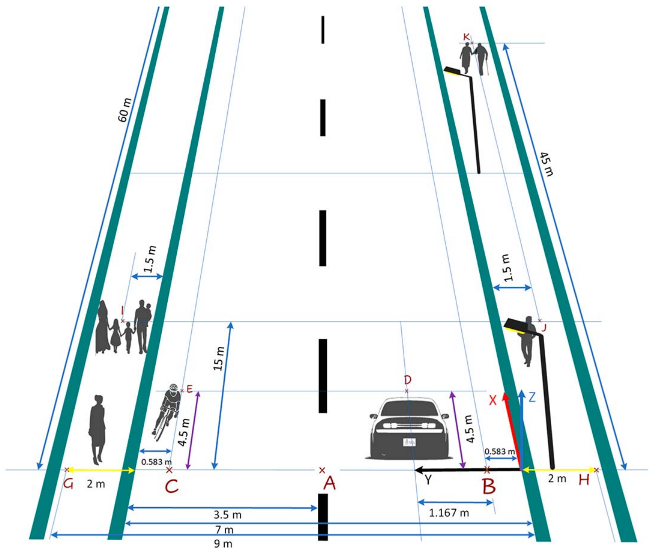

Fig. 1. Observed POI A, B, . . ., K.

$$\gamma = \arccos\left(\frac{h\cos\sigma + (y-d)\sin\sigma}{[(y-d)^2 + x^2 + h^2]^{1/2}}\right) \quad (3)$$

where σ denotes the inclination of the boom.

These relations quantify each luminaire's contribution to specific street point p depending only on its coordinates and luminaire geometry, thus providing the mathematical model for the calculation of exact lighting adjustments based on location characteristics and orientation.

Fig. 1 depicts the street geometry and distances (coordinates) of observed points of interest (POI) A, B, C, D, E, G, H, I, J, and K, for which illuminance E_p is observed. Note that not all of the POI are on the road surface and some correspond to the observers eye height. Section of the road between two luminaires is treated as a single street segment, and each POI is contributed by only four luminaires.

The distance between luminaires is 30 m, lamp height is $h = 11$ m, light point overhang is 0.1 m, boom inclination is $0°$, and boom length is $d = 0.1$ m. The parameters are provided by industry partner.

By applying (1), the illuminance intensity distribution E_p at each POI is computed as a function of the luminaire's luminous intensity I, i.e., $E_p = f(I)$. Full model of a single street segment including 10 POIs is

$$\begin{bmatrix} E_A \\ E_B \\ E_C \\ E_D \\ E_E \\ E_G \\ E_H \\ E_I \\ E_J \\ E_K \end{bmatrix} = \begin{bmatrix} 0.0072 & 0.0003 & 0.0003 & 0.0003 \\ 0.0082 & 0.0003 & 0.0003 & 0.0000 \\ 0.0054 & 0.0003 & 0.0003 & 0.0000 \\ 0.0085 & 0.0005 & 0.0002 & 0.0001 \\ 0.0060 & 0.0005 & 0.0002 & 0.0001 \\ 0.0039 & 0.0003 & 0.0003 & 0.0000 \\ 0.0078 & 0.0003 & 0.0003 & 0.0003 \\ 0.0014 & 0.0014 & 0.0001 & 0.0001 \\ 0.0020 & 0.0020 & 0.0001 & 0.0001 \\ 0.0001 & 0.0019 & 0.0000 & 0.0019 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix}. \quad (4)$$

3) Model Validation by Ray Tracing Software: The relation (1) and its elaborated form (4) are further applied on a street segment of 10×6 symmetrically distributed points on the road surface as a typical approach in lighting design process, used both in street lighting norms and planning projects. The outputs of the process are illuminance values listed in Table II. First three rows (y_1, y_2, y_3) correspond to left lane and last three rows (y_4, y_5, y_6) to the right lane.

To verify the model, the considered street is also configured in COMSOL Multiphysics software with the Ray Optics Module that simulates tracing ray trajectories to project light distribution from each luminaire onto the road surface, as shown in Fig. 2. The simulation measured irradiance $(\mathrm{W/m^2})$, linearly correlated with luminous intensity in the visible spectrum, enabling precise illuminance calculations at specific points. These results validate the model's accuracy and its potential to enhance predictive control algorithms for LED street lighting. They also validate the utilization of professional software tools in street lighting design.

Model outputs from Table II are additionally validated by a professional lighting ray tracing software DIALux, with outputs as shown in Fig. 3. The verification indicates high accuracy,

TABLE II
ILLUMINANCE VALUES CALCULATED FROM MODEL

E_p (lx)	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
y_1	26	25	22	20	17	17	20	22	25	26
y_2	29	28	24	20	18	18	20	24	28	29
y_3	33	30	26	20	18	18	20	26	30	33
y_4	34	31	26	20	17	17	20	26	31	34
y_5	36	32	26	20	16	16	20	26	32	36
y_6	36	31	25	18	15	15	18	25	31	36

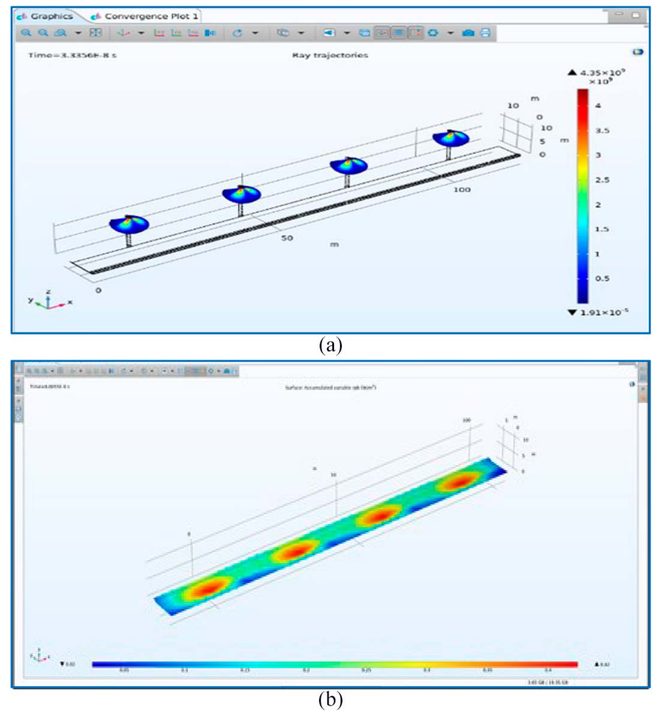

Fig. 2. Computed COMSOL ray tracing model: (a) luminaire intensity distribution; and (b) road surface intensity distribution.

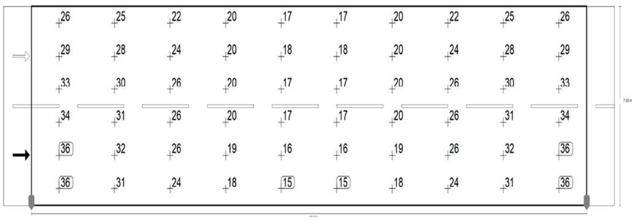

Fig. 3. Road illuminance obtained by a professional lighting software.

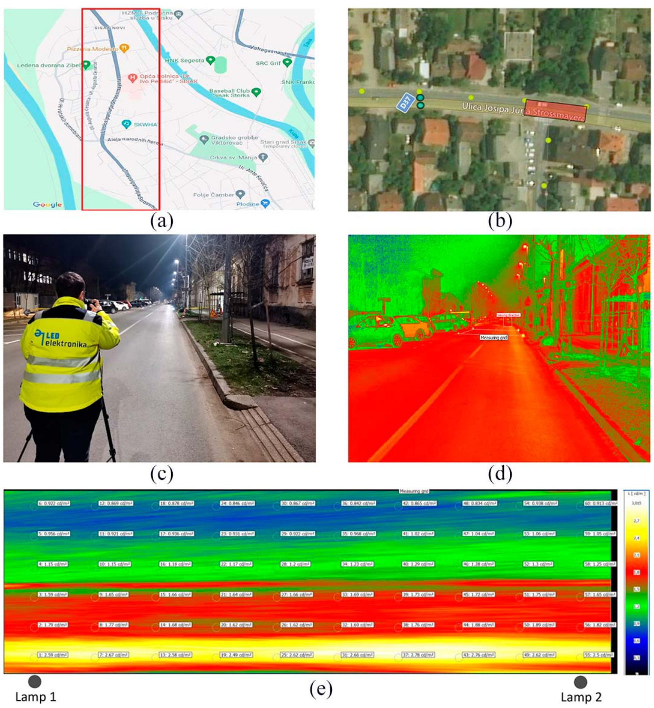

Fig. 4. Illuminance measurement at J. J. Strossmayer: (a) reference surface location; (b) reference surface view; (c) lighting technology measurement; (d) luminance photo; and (e) street measurements obtained from the right side.

with a deviation of less than 0.4% for the average error of 60 points across four luminaires, with details presented in [37].

B. Experimental Verification of the Model

Comprehensive measurements of luminance and geometry were conducted on the real considered street. Fig. 4 presents the illuminance measurements process, highlighting the reference surface location and view, and the technologies used for lighting intensity measurement. Fig. 4(e) shows the luminance measurements (in cd/m^2) obtained by luminance imaging camera. The measurement values are additionally presented in Table III.

The process implies an observer [camera, Fig. 4(c)], placed at the height of 1.5 and 60 m from the first luminaire on the observed segment. Camera images are captured both from the left and right sides of the road, and measurements [Fig. 4(e)] are extracted [Fig. 4(d)]. The same process is conducted as validation and certification of street lighting installation, occasional audit, and are also such defined in the street lighting norms.

TABLE III
MEASUREMENTS OBTAINED BY LUMINANCE IMAGING CAMERA

L	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
y_1	0.922	0.869	0.878	0.846	0.867	0.842	0.865	0.834	0.938	0.913
y_2	0.956	0.921	0.936	0.931	0.922	0.968	1.02	1.04	1.06	1.05
y_3	1.15	1.15	1.18	1.17	1.2	1.23	1.29	1.28	1.3	1.25
y_4	1.59	1.65	1.66	1.64	1.66	1.69	1.73	1.72	1.75	1.65
y_5	1.79	1.77	1.68	1.62	1.62	1.69	1.76	1.88	1.89	1.82
y_6	2.59	2.67	2.58	2.49	2.62	2.66	2.78	2.76	2.5	2.5

TABLE IV
REFLECTANCE FACTOR, DIALUX SOFTWARE

R	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
y_1	0.133	0.138	0.157	0.157	0.185	0.185	0.173	0.157	0.138	0.133
y_2	0.141	0.146	0.157	0.188	0.192	0.209	0.188	0.17	0.146	0.141
y_3	0.143	0.157	0.169	0.204	0.24	0.24	0.22	0.193	0.157	0.143
y_4	0.166	0.182	0.193	0.236	0.277	0.277	0.251	0.217	0.182	0.166
y_5	0.183	0.196	0.23	0.281	0.334	0.353	0.314	0.242	0.206	0.183
y_6	0.192	0.213	0.249	0.297	0.356	0.377	0.332	0.275	0.213	0.192

In the rest of the article [and in Fig. 4(e)], we consider only observations taken from the right side of the road.

To verify the derived model from (1), camera measurements are now compared with model outputs. However, the model provides illuminance E_p (in lx) while camera gives the luminance L (in cd/m^2) as a reflection of the road in the eye of the 60 m distant observer. For this, we relate the two by taking into account the road reflectance R as

$$E_p = \frac{L\pi}{R}. \tag{5}$$

In lighting systems, reflectance R is either experimentally or, more often, empirically determined. It changes with the road conditions such as humidity or simply over time by asphalt aging. For the purpose of determining the reflectance R, we utilize DIALux, a professional lighting design software. For the same street configuration as considered before, the software outputs the road reflectance for the considered road segment as given in Table IV.

If reflectance values from Table IV are applied directly to illuminance model outputs from Table II, an equivalent luminance table is obtained (in cd/m^2) that can be directly compared to the obtained experimental camera measurements. This results in 4.3707% of mean percentage error (MPE), but 17.3463% root mean squared percentage error (RMSPE) as discrepancy varies both to positive and negative sides. The difference between the two we tribute to the real-world street topology and testing condition.

The most expressed factor we observed is the road curvature where the middle of the road is higher than end sides as the inherent drainage system. This curvature directly and significantly influences the reflectance factor (as expected). To respect this, a simple correction of downscaling left lane ($y_{1,2,3}$) reflectance by 20% and upscaling 10% for right lane ($y_{4,5,6}$), is introduced, which is in line for the right-side observations. With this, the MPE and RMSPE of the model, respectively, become 1.7111% and 10.6132%.

The remaining discrepancy is due to installation imperfections of street topology such as slight deviations of light source

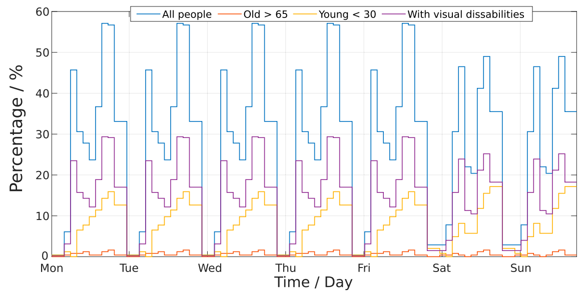

Fig. 5. Percentage of surveyed people present at different times on city streets.

inclination, road width, distance between luminaires, nonhomogeneous aging of the asphalt, etc., and due to imperfections of the measurement process such as ambient light contribution, i.e., reflection from nearby buildings and parked cars.

With this verification, the model is experimentally validated and further used in the predictive controller design and in energy efficiency cost-benefit analysis.

C. Microlocation Data

The proposed control system dynamically adapts to environmental and urban changes, optimizing lighting and conserving energy considering factors such as weather, traffic, pedestrian flow, and street characteristics.

1) Data Sources: A robust data framework integrates.

a) *Weather data*: Historical data and forecasts on temperature (°), humidity (%), visibility (m), and precipitation (mm/m^2), obtained from nearby on-site measurements and Open Weather forecasts.

b) *Pedestrian data*: Pedestrian profiles by location, and demographic features obtained from a survey conducted among 245 participants and expressed in percentage as shown in Fig. 5.

c) *Traffic data*: Vehicle flow, congestion, and peak traffic, quantified by ”Jam Factor” (JF).

i) JF 0–4: Light traffic.

ii) JF 4–8: Moderate to slow traffic.

iii) JF 8–10: Heavy traffic, (JF 10 indicating blockage).

d) *Street details*: Street geometry and point-in-space lighting priorities and needs obtained from experience of industry partner.

Jointly, the data framework to describe current and future microlocation conditions on the road is referred to as WPTS. For our case study, we have acquired on-site correlated data from mid February 2021 to April 2022.

2) Data Processing and Model Application: Meteorological variables in weather data can be obtained from official forecasting services or open data sources. These are typically numerical weather predictions (NWPs) from physics-based models for discrete atmospheric blocks (e.g., 4×4 km). To create hyperlocal forecasts, we pair NWP with historical on-site measurements using machine learning. Specifically, we fuze local road weather sensor data with open weather service data using a TabNET artificial neural network. This improves prediction

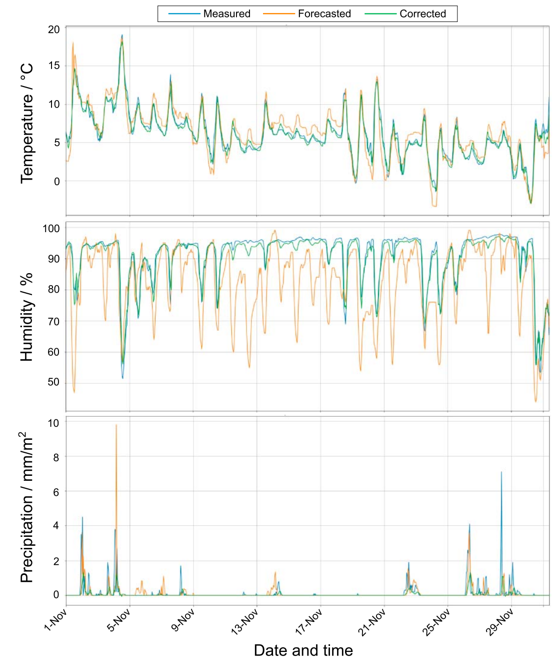

Fig. 6. Precipitation, temperature, and humidity forecasts (recorded, forecasted, and corrected).

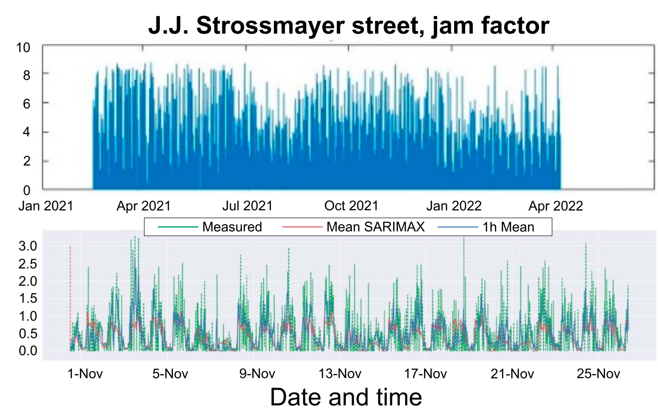

Fig. 7. Traffic data at J.J. Strossmayer street.

accuracy, reducing temperature RMSE from 2.46 °C to 2.11 °C and relative humidity error from 12.01% to 7.91% (example: 24 h dataset). Results for the entire year are shown in Fig. 6.

Traffic data is obtained as floating car data (FCD) from a navigational service, based on mobile phone and GPS data. Data for J.J. Strossmayer street is shown in Fig. 7. More users provide more accurate traffic flow information. Predictions are available from navigational services or can be modeled from historical data [38]. For street lighting, we use the SARIMAX model applied to historical data. Predictions in Fig. 7 show November 2021 results alongside real FCD.

The data determines dynamic street conditions using widely available sources. It can be extended with specialized sensors such as intersection cameras for precise vehicle counts and speeds.

Lighting optimization uses WPTS data to generate data-driven models and calculate street condition predictions for MPC [39]. Weather and traffic predictions are obtained via

TABLE V
ADAPTIVE DIMMING PROFILE EXAMPLE

	Road Conditions	Thresholds	Dimming Profile
1	Clear, dry Light traffic Few pedestrians	Precipitation = 0 Visibility $\geq$ 10 000 Traffic density $<$ 4 Pedestrians total $<$ 10.0	**D1** 0.3@80% 0.7@20%
2	Clear, dry Moderate traffic Moderate pedestrians	Precipitation = 0 Visibility $\geq$ 10 000 4 $\leq$ traffic density $<$ 8 10.0 $\leq$ pedestrian total $\leq$ 50.0	**D2** 0.3@90% 0.7@20%
3	Clear, dry Severe traffic Many pedestrians	Precipitation = 0 Visibility $\geq$ 10 000 Traffic density $>$ 8 Pedestrians total $>$ 50.0	**D3** 0.3@100% 0.7@30%
4	Rain Moderate traffic Moderate pedestrians	Precipitation $>$ 0 Humidity $\geq$ 90 Visibility $\leq$ 10 000 4 $\leq$ traffic density $<$ 8 10.0 $\leq$ pedestrian total $\leq$ 50.0	**D4** 0.3@100% 0.7@40%
5	Snow Moderate traffic Moderate pedestrians	Precipitation $>$ 0 Humidity $\geq$ 90 Temperature $<$ 0 Visibility $\leq$ 10 000 4 $\leq$ traffic density $<$ 8 10.0 $\leq$ pedestrian total $\leq$ 50.0	**D5** 0.3@80% 0.7@40%

TABLE VI
ILLUMINANCE INTENSITIES AT POI

Points of Interest	A	B	C	D	E	G	H	I	J	K
Illuminance (lx)	32	36	18	33	19	5	18	8	7	6

machine learning (ML)-based algorithms, while pedestrian data comes from surveys. These predictions are transformed into individual dimming profiles for each lamp based on scenarios and thresholds in Table V.

The dimming profile uses duration-power notation (e.g., 0.3@80% for 80% power over 30% of night-operation time). Traditionally set once during commissioning, our approach provides real-time updated profiles over a prediction horizon. Critical locations (intersections, crossings, and schools) receive increased power, fully detailing the micro-location concept.

3) Dynamic Reference Generation: The proposed approach integrates real-time data—including weather conditions, pedestrian diversity, traffic patterns, and street conditions to generate predictive data for a horizon N with a time resolution T_s. The resulting predictive data is dynamically fed into the proposed model to generate adaptive lighting profiles, and thereby generating the dynamic reference for the MPC algorithm. Importantly, these profiles are applied within standard time slots, ensuring compliance with regulations while optimizing lighting based on real-time and predictive insights.

D. MPC Design

1) Mathematical Model and Framework for Dynamic Lighting: The MPC algorithm for street lighting dynamically adjusts illumination based on temporal changes and situational needs, optimizing lighting delivery and transitions within a predictive horizon. These adjustments, predefined by authorities or automated, adapt to the weather, pedestrians, traffic, and street topology, ensuring visibility and energy efficiency. In general, the MPC framework uses the state-space model, generally formalized as

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k \\ y_k &= Cx_k \end{aligned} \tag{6}$$

where

a) k: Discrete-time step index.
b) $x_k \in \mathbb{R}^m$: State vector of illuminance intensities at m number of observed POI, $x_k = [E_1, E_2, \ldots, E_m]^T$.
c) $u_k \in \mathbb{R}^n$: Input vector of light intensities $u_k = [i_1, i_2, \ldots, i_n]^T$ for n luminaires expressed in percentage as of rated values curve from datasheet polar diagram.
d) $y_k \in \mathbb{R}^m$: Output vector representing delivered illuminance.

In matrix form, (6) is

$$\underbrace{\begin{bmatrix} E_A(k)\\ E_B(k)\\ E_C(k)\\ E_D(k)\\ E_E(k)\\ E_G(k)\\ E_H(k)\\ E_I(k)\\ E_J(k)\\ E_K(k)\end{bmatrix}}_{\boldsymbol{x}_{k+1}} = \underbrace{\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\end{bmatrix}}_{\boldsymbol{A}}$$
$$+ \underbrace{\begin{bmatrix} E_A(k)\\ E_B(k)\\ E_C(k)\\ E_D(k)\\ E_E(k)\\ E_G(k)\\ E_H(k)\\ E_I(k)\\ E_J(k)\\ E_K(k)\end{bmatrix}}_{\boldsymbol{x}_k} + \underbrace{\begin{bmatrix} 0.0072 & 0.0003 & 0.0003 & 0.0003\\ 0.0082 & 0.0003 & 0.0003 & 0.0003\\ 0.0054 & 0.0005 & 0.0003 & 0.0000\\ 0.0085 & 0.0005 & 0.0002 & 0.0001\\ 0.0060 & 0.0005 & 0.0002 & 0.0001\\ 0.0039 & 0.0003 & 0.0003 & 0.0001\\ 0.0078 & 0.0003 & 0.0003 & 0.0003\\ 0.0014 & 0.0020 & 0.0020 & 0.0020\\ 0.0020 & 0.0020 & 0.0020 & 0.0019\\ 0.0001 & 0.0019 & 0.0000 & 0.0019\end{bmatrix}}_{\boldsymbol{B}} \underbrace{\begin{bmatrix} i_1\\ i_2\\ i_3\\ i_4\end{bmatrix}}_{\boldsymbol{u}_k} \tag{7}$$

$$\underbrace{\begin{bmatrix} E_A(k+1)\\ E_B(k+1)\\ E_C(k+1)\\ E_D(k+1)\\ E_E(k+1)\\ E_G(k+1)\\ E_H(k+1)\\ E_I(k+1)\\ E_J(k+1)\\ E_K(k+1)\end{bmatrix}}_{\boldsymbol{y}_k} = \underbrace{\begin{bmatrix} 1&0&0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0&0&0\\ 0&0&0&0&1&0&0&0&0&0\\ 0&0&0&0&0&1&0&0&0&0\\ 0&0&0&0&0&0&1&0&0&0\\ 0&0&0&0&0&0&0&1&0&0\\ 0&0&0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&0&0&1\end{bmatrix}}_{\boldsymbol{C}} \underbrace{\begin{bmatrix} E_A(k)\\ E_B(k)\\ E_C(k)\\ E_D(k)\\ E_E(k)\\ E_G(k)\\ E_H(k)\\ E_I(k)\\ E_J(k)\\ E_K(k)\end{bmatrix}}_{\boldsymbol{x}_k}. \tag{8}$$

Since the light propagation is considered instantaneous, the model is static, implying $A = 0$. However, we define MPC to impose gradual transitions by restricting the rate of change of control variables. As such, input values from the previous step, u_{k-1} are included in the state vector and thus the state matrix $A \neq 0$. This is described in detail in [3].

The street is parted into segments, each incorporating four luminaires to illuminate POI (A–H) as depicted in Fig. 1. For a single road segment $n_{s,t}$, the model is

$$x_{n_{s,t},k+1} = A_{n_{s,t}} x_{n_{s,t},k} + B_{n_{s,t}} u_{n_{s,t},k} \tag{9}$$

with $u_{n_{s,t}} = [i_{n_{s,t-1}}, i_{n_{s,t}}, i_{n_{s,t+1}}, i_{n_{s,t+2}}]^T$, where $i_{n_{s,t}}$ is the closest, dominant luminaire, preceding one is $i_{n_{s,t-1}}$, and two following ones as $i_{n_{s,t+1}}$ and $i_{n_{s,t+2}}$. Such segments are arbitrarily stacked into a street configuration of n luminaires and m POI. This way, the illumination of POI (A–H) is consecutively extended for each luminaire thus forming longitudinal street illumination zones. As some of the points have vertical component (coordinate $z \neq 0$), zones are also in 3-D space.

2) Objective Function: The optimization minimizes deviations between current illuminance in j point of interest $x_{j,k}$ and desired levels $x_{j,k}^{\text{ref}}$ over the prediction horizon N, in context of tradeoff with energy efficiency

$$\min_{u_{i,k}} \sum_{k=1}^{N} \sum_{j=1}^{m} \sum_{i=1}^{n} w_{x,j,k} \left| x_{j,k} - x_{j,k}^{\text{ref}} \right| + w_{u,i,k} u_{i,k} \tag{10}$$

subject to

a) Strict norms: $x_{j,k} \geq x_{\min}$ for points $j \in S$
b) State update: $x_{x+1} = Ax_k + Bu_k$
c) Output: $y_k = Cx_k$
d) Control bounds:

$$0 \leq u_k \leq u_{\max}$$
$$\Delta u_{\min} \leq u_{k+1} - u_k \leq \Delta u_{\max}$$

Variables and parameters include.

a) $x_{j,k}^{\text{ref}}$: Desired illuminance at point of interest j derived from micro-location WPTS.
b) w: Weighting coefficients balancing energy and illuminance priorities.
c) S: Points with strict quality norms.
d) $u_{\max}$, $\Delta u_{\min}$, and $\Delta u_{\max}$: Control constraints.

The approach adjusts lighting using real-time data, predictive analytics, and predefined standards to optimize illumination and energy efficiency.

III. Case Study and Results

A. Street Lighting MPC

The proposed method was tested at priority observation points representing various road users (vehicles, cyclists, and pedestrians) as shown in Fig. 5. The goal was to dynamically adjust the illuminance based on environmental conditions, ensuring energy efficiency without disrupting traffic. The target illuminance levels were derived from real measurements, summarized in scenarios, and given in Table VI. Fig. 8 shows the flowchart of the proposed algorithm in the simulation environment.

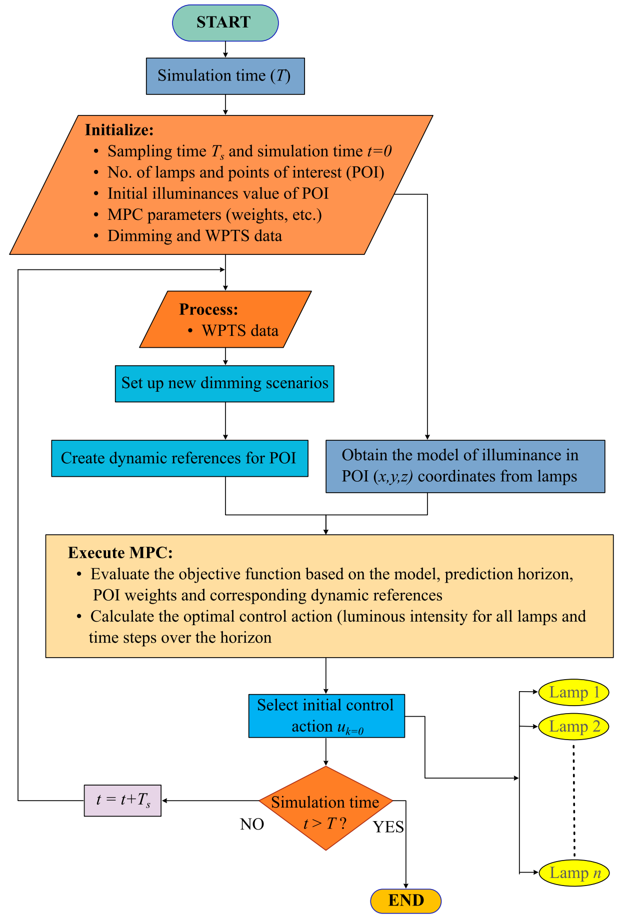

Fig. 8. Flowchart of the proposed algorithm.

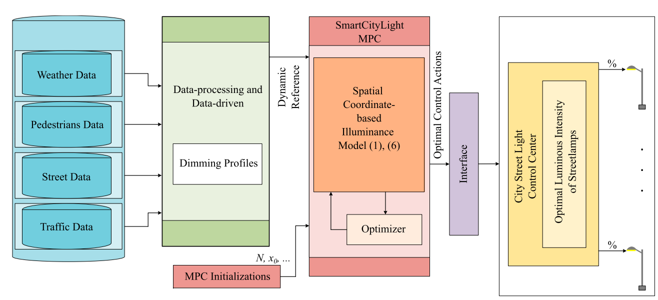

Fig. 9. Block diagram of the environment simulation model.

The control algorithm in each time step k includes.

1) Obtaining the latest WPTS data.
2) Updating data-based models and future profiles of WPTS.
3) Calculation of reference illuminance based on WPTS data.
4) Calculating control actions on the prediction horizon N.
5) Application of current control actions to luminaires.

Fig. 9 illustrates a block diagram of the environment simulation model for street lighting optimization. It includes a database storing data such as dimming profiles and traffic information. The data-processing module generates adaptive dimming scenarios for the “SmartCityLight-MPC” module, which

TABLE VII
MEASURED ILLUMINANCE VALUES ON THE SELECTED STREET

Symbol	Average (*lx*)	Minimum (*lx*)	Maximum (*lx*)
E	24.6	15.1	36

TABLE VIII
MEASURED LUMINANCE COMPARED TO TARGET VALUES

Symbol	Calculated $(\mathrm{cd/m^2})$	Target $(\mathrm{cd/m^2})$
L_{av}	1.55	$\geq$ 1.00

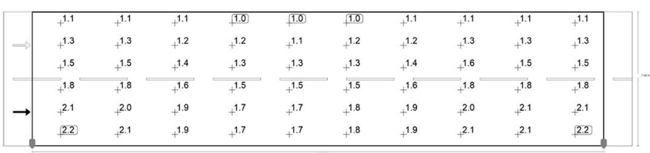

Fig. 10. Targeted luminance levels at measured points.

calculates optimal illuminance levels. The "Optimizer" determines control actions relayed through an Interface to the "City Street Light Control Center," ensuring energy-efficient street lighting based on real-time conditions.

In accordance with the adopted EIC standards for the selected street, the measured illuminance (E) values, including average, minimum, and maximum levels in lux (lx), are presented in Table VII.

Furthermore, the measured minimum luminance (L_{av}) in $\mathrm{cd/m^2}$ is evaluated and summarized in Table VIII. The obtained values meet the minimum required standards, ensuring adequate visibility and safety.

Fig. 10 illustrates the targeted luminance levels at the ten designated POI on the selected street. These measurements confirm compliance with the corresponding standards, ensuring that the street lighting system effectively addresses safety concerns under varying conditions.

By meeting the minimum illuminance levels prescribed by the standards, the lighting design ensures adequate visibility for pedestrians and drivers, contributing to overall road safety.

B. Results

To evaluate the MPC-based street lighting control system, a MATLAB/Simulink model is developed for J. J. Strossmayer Street in Sisak, Croatia. The proposed simulation model integrates an MPC framework to optimize the luminous intensity of streetlights based on real-time environmental conditions. The system receives multiple input parameters, including original dimming profiles, pedestrian diversity profile, weather conditions, and road surface status. These inputs are processed to generate an adaptive dimming strategy that ensures optimal illumination while minimizing energy consumption. At the core of the MPC controller is the model used as a basis to dynamically adjust the lighting intensity by predicting future states and optimizing control actions accordingly. The controller receives dimming profiles and computes an optimal response that regulates the streetlight outputs. The predicted luminous intensity is continuously updated based on real-time data, ensuring that lighting conditions remain adaptive to varying environmental and pedestrian factors. The output control signals can be transmitted to the street lighting infrastructure, where they can be executed to enhance visibility and safety. The simulation features a 5-min sampling interval and a 6-h prediction horizon ($N = 72$). Control input constraints are $\Delta u_{\min} = -0.05$ and $\Delta u_{\max} = 0.05$ per interval as $\pm 5\%$ allowed rate of change, corresponding to about ± 250 cd (depending on POI location). The algorithm is tested over one year while adjusting the illuminance at the points $\{A, B, C, D, E, G, H, I, J, K\}$ according to their priorities. The consumption is further summed for all luminaires over the simulation period, and compared by using the original dimming profile currently present on the streets.

C. Energy Efficiency

The total energy consumption for a single luminaire over a one-year simulation with a 5-min sampling time is calculated as

$$E_n = \sum_{k=1}^{365\times 24\times 12} P \times u_k \times T_s \tag{11}$$

for both the current and optimized control variables, denoted as E_0 and E^*, respectively. Here, $P = 120$ W represents the nominal power of the existing luminaire, which scaled by the control variable u_k in each time step over a total simulation duration of one year, i.e. $365 \times 24 \times 12$ intervals of 5 min, T_s. The control variable u_k is defined as

$$u_k = \begin{cases} u_{ko}, & \text{for } E_0 \\ u_k^*, & \text{for } E^* \end{cases} \tag{12}$$

where u_{ko} represents the control variable of the original dimming profile as a percentage of the luminaire's input power, while u_k^* corresponds to the control variable of the adaptive dimming profile optimized by the MPC algorithm, as detailed in Table V.

1) Case 1: Original Dimming Profile: The existing dimming profile for J. J. Strossmayer Street follows a fixed schedule of 0.5@100% and 0.5@50%, meaning the luminaires operate at full power (100%) for half the time and at reduced power (50%) for the remaining period. The one-year simulation results, along with a zoomed-in view of a selected day (13 December 2021), are presented in Fig. 11, showcasing illuminance intensity variations at key points.

The total energy consumption for this profile is $E_0 = 186.86$ kWh.

2) Case 2: Adaptive Dimming Profile: The implementation of the optimized dimming profile, generated by the MPC algorithm, resulted in a significant reduction in energy consumption. The total energy usage for this scenario is $E^* = 127.19$ kWh.

The corresponding simulation results are shown in Fig. 12, while Fig. 13 illustrates the MPC-based control actions for dynamic dimming adjustments for a single street segment and POI

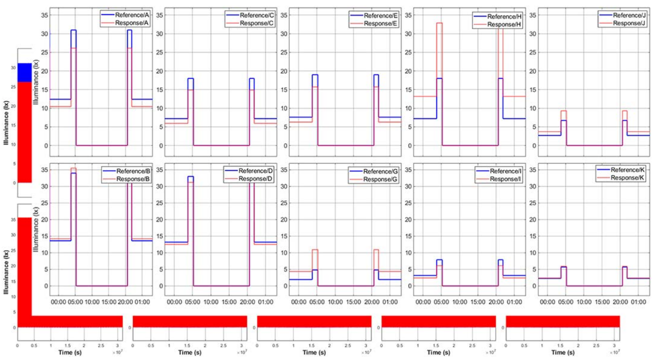

Fig. 11. Illuminance intensity at selected points using the proposed MPC algorithm with the original dimming profile (one-year simulation, zoomed-in on 13 December 2021).

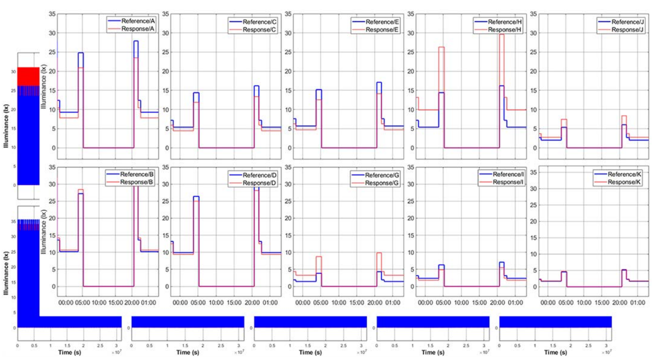

Fig. 12. Illuminance intensity at selected points using the proposed MPC algorithm with the adaptive dimming profile (one-year simulation, zoomed-in on 13 December 2021).

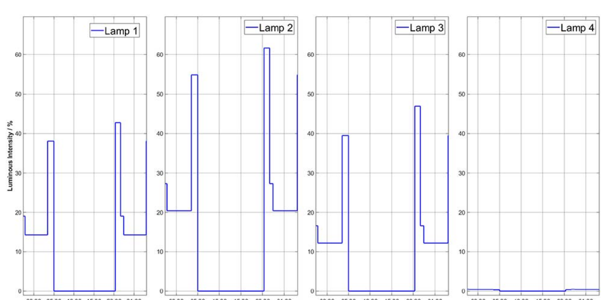

Fig. 13. MPC control actions (one-year simulation, zoomed-in on 13 December 2021).

positioned under luminaire 2, without gradual transition constraints ($\Delta u_{\min} = \Delta u_{\max} = 0$). Control variables show how adjacent lamps contribute to the POI of focus to deliver optimal tradeoff in 3-D space.

3) Annual Energy Savings Analysis: The simulations are conducted using WPTS data from May 2021 to April 2022. The daily energy consumption of a single luminaire throughout the year is depicted in Fig. 14, highlighting seasonal variations due to changing night durations. The MPC algorithm dynamically reduces lighting levels to the minimum of 30% in the absence of pedestrians or vehicles, accumulating significant energy savings over the year.

Energy use with MPC varies throughout the year, reflecting changes in WPTS data and the scenarios in Table V.

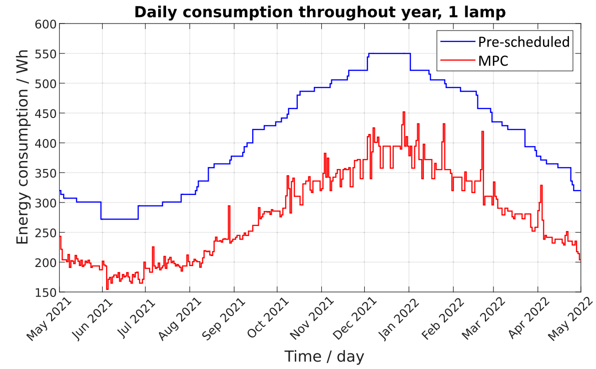

Fig. 14. Daily energy consumption of a single luminaire over one year using prescheduled and MPC-optimized dimming strategies.

For the current system, one luminaire consumes 186.86 kWh annually, while the entire street with 100 luminaires consumes 18.686 MWh. With the MPC method, consumption drops to 127.19 kWh for one luminaire and 13.705 MWh for the 100-luminaire street, representing a 31.94% reduction for one luminaire and 26.65% for the street. The variation in reduction rates is due to safety-critical areas, such as pedestrian crossings and intersections. The results therefore clearly indicate that the MPC approach, with its adaptive dimming strategy, significantly reduces energy consumption compared to the current system.

D. Comparison to Literature

The proposed MPC framework contributes beyond the current state of the art as listed in Table I. While prior systems, such as [4]'s IR/PIR sensors enable only reactive vehicle detection and [5]'s movement-based dimming operates at fixed levels without weather adaptation, our MPC achieves dynamic, multiobjective optimization. The comparison further differentiates our work from fuzzy neural networks [7], which rely on heuristic rules; multisensory IoT systems [8], limited by computational costs; and static dimming [28], constrained by inflexible scheduling.

Authors in [28] use traffic flow predictions and dim down the lighting to 50% when it is not needed, ensuring savings of 22% by adaptive dimming, compared to static dimming. Authors in [27] report energy savings of 45%–98% in extreme cases of complete switch off compared to constant operation at full power for 112 streetlights and one week of operation, which makes direct comparison impossible. As an indirect comparison, our approach saves 50.98% when related to constant operation at full power, but it does not completely switch off the lamp when not needed. To the best of our knowledge, there is no MPC application or ray-tracing model-based optimization in city lighting, which is also confirmed by patent [39].

E. Implementation Aspects

Basic prerequisite to implement the technology is ability of the luminaires to exchange information with remote server and accept control signals. Today there are controllers based on IoT

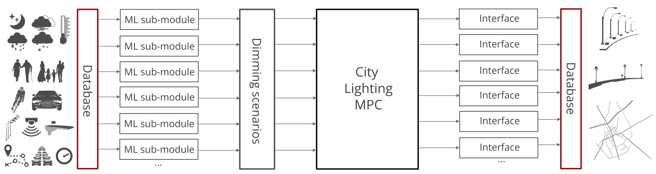

Fig. 15. Control scheme.

concept that exploit ZHAGA standardized interface available in all modern LED street lamps. With retail cost in the rank of 20$ intended to retrofit existing lamps with boom lift, they offer individual dimming capability to be fully exploited by the proposed MPC algorithm.

The control algorithm steps from Section III-A are illustrated in control scheme block diagram in Fig. 15.

Usual practice with LED lighting is a street-controller that jointly controls a larger number of lamps, and can be similarly upgraded to enable information exchange with a remote server as a significantly cheaper solution but for street-level dimming. Most of the technology is already a standard today, requiring thus only access to an established database. Algorithmwise, the costs are reduced to data-driven modeling and MPC solvers, for which there are also open source editions available. Critical part is an implementation project of transforming street geometry to formulate MPC problem and implementation of a software architecture.

IV. Conclusion

This article presents a cost-effective urban street lighting modernization strategy that integrates energy-efficient LED technology with an advanced MPC approach. The system achieves up to 31.94% energy savings per luminaire while adhering to safety standards, leveraging adaptive dimming and real-time microlocation data to dynamically adjust to environmental conditions. However, the study has some limitations, including its partial reliance on simulated traffic and pedestrian data, which may not fully capture real-world variability, as well as potential computational challenges in scaling MPC optimization for large deployments. In addition, the analysis prioritizes energy efficiency over long-term maintenance and reliability considerations. Future research should expand into real-world field trials across diverse urban settings to validate robustness, while also exploring more efficient MPC algorithms to enhance scalability. Hybrid control strategies combining MPC with complementary methods could further improve flexibility and provide a broader evaluation framework. Despite these areas for refinement, the study establishes MPC as a promising approach for smart city lighting, balancing energy savings with performance.

References

[1] C. C. on Energy Efficiency, "Public lighting toolbox for municipal energy efficiency implementation," 2024. Accessed: Oct. 4, 2024. [Online]. Available: https://c2e2.unepccc.org/collection/public-lighting-toolbox-for-municipal-energy-efficiency-implementation/

[2] F. Marino, F. Leccese, and S. Pizzuti, "Adaptive street lighting predictive control," in *Proc. 8th Int. Conf. Sustain. Energy Build. (SEB-16)*, Turin, Italy, Sep. 2016. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1876610217302710

[3] H. I. Shaheen, M. Gapit, A. Martinović, and V. Lešić, "Street lighting optimal dimming with model predictive control," in *Proc. 62nd IEEE Conf. Decis. Control (CDC)*, 2023, pp. 8864–8869.

[4] A. M. Al-Smadi, S. T. Salah, A. A. Al-Moomani, and M. S. Al-Bataineh, "Street lighting energy-saving system," in *Proc. 16th Int. Multi-Conf. Syst., Signals Devices (SSD)*, 2019, pp. 763–766.

[5] S. Dhamodaran, T. Kailash, G. T. Srinivasan, and M. Lakshmi, "Dynamic control of street lights with human mobility, illumination," in *Proc. Int. Conf. Comput. Power, Energy Inf. Commun. (ICCPEIC)*, 2017, pp. 554–557.

[6] C.-W. Yang, E. Nefedov, S. Sierla, and P. Flikkema, "Vehicle and pedestrian aware street lighting automation," in *Proc. IEEE 13th Int. Conf. Ind. Informat. (INDIN)*, 2015, pp. 1269–1274.

[7] Y. Xue, Y. Zhang, and P. Shuxue, "An intelligent dimming algorithm of streetlight based on fuzzy neural network," in *Proc. Int. Wireless Commun. Mobile Comput. (IWCMC)*, 2020, pp. 1964–1967.

[8] H. B. Khalil, N. Abas, and S. Rauf, "Intelligent street light system in context of smart grid," in *Proc. 8th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, 2017, pp. 1–5.

[9] D. Saha, S. M. Sorif, and P. Dutta, "Weather adaptive intelligent street lighting system with automatic fault management using Boltuino platform," in *Proc. Int. Conf. ICT Smart Soc. (ICISS)*, 2021, pp. 1–6.

[10] S. Słomiński, "Advanced luminance modeling of light sources for simulation and computational purposes of lighting parameters," in *Proc. 7th Lighting Conf. Visegrad Countries (Lumen V4)*, 2018, pp. 1–5.

[11] X.-H. Lee, I. Moreno, and C.-C. Sun, "High-performance led street lighting using microlens arrays," *Opt. Exp.*, vol. 21, no. 9, pp. 10612–10621, May 2013.

[12] D. Garcia-Llera et al., "Optimizing led lamps design for street lighting with staggered arrangement allowing energy saving strategies in a lighting smart grid context," in *Proc. IEEE Ind. Appl. Soc. Annu. Meeting*, 2015, pp. 1–8.

[13] V. Muzet et al., "Review of road surface photometry methods and devices—Proposal for new measurement geometries," *Lighting Res. Technol.*, vol. 53, no. 3, pp. 213–229, 2021.

[14] M. Dwiyaniti et al., "Energy saving of IoT-based light intensity on smart streetlight," in *Proc. 5th Int. Conf. Comput. Informat. Eng. (IC2IE)*, 2022, pp. 6–9.

[15] S. M. Sorif, D. Saha, and P. Dutta, "Smart street light management system with automatic brightness adjustment using bolt IoT platform," in *Proc. IEEE Int. IoT, Electron. Mechatron. Conf. (IEMTRONICS*), 2021, pp. 1–6.

[16] F. Agramelal, M. Sadik, A. E. Hannani, and Y. Moubarak, "A traffic-aware street lighting system based on fuzzy logic controller," in *Proc. IEEE 18th Int. Colloq. Signal Process. IEEE Int. Symp. Spread Spectr. Tech. Appl. (CSPA)*, 2022, pp. 132–137.

[17] Y. Wang, Z. Deng, K. Tao, T. Liu, X. Bao, and B. Cheng, "Intelligent urban street lighting system based on IoT cloud platform," in *Proc. 9th Int. Conf. Dependable Syst. Their Appl. (DSA),* 2022, pp. 192–197.

[18] H. Sun, Y. Chen, L. Wang, J. Song, J. Sun, and X. Liu, "Design of smart street light management system based on Internet of Things," in *Proc. IEEE 6th Adv. Inf. Technol., Electron. Automat. Control Conf. (IAEAC)*, 2022, pp. 998–1003.

[19] C. Jing, D. Shu, and D. Gu, "Design of streetlight monitoring and control system based on wireless sensor networks," in *Proc. 2nd IEEE Conf. Ind. Electron. IEEE Int. Symp. Spread Spectr. Tech. Appl.*, 2007, pp. 57–62.

[20] A. Abdullah, S. H. Yusoff, S. A. Zaini, N. S. Midi, and S. Y. Mohamad, "Smart street light using intensity controller," in *Proc. 7th Int. Conf. Comput. Commun. Eng. (ICCCE)*, 2018, pp. 1–5.

[21] D. Sunehra and S. Rajasri, "Automatic street light control system using wireless sensor networks," in *Proc. IEEE Int. Conf. Power, Control, Signals Instrum. Eng. (ICPCSI)*, 2017, pp. 2915–2919.

[22] A. R. Alba and N. B. Linsangan, "Application specific computer for intelligent monitoring of cabled streetlights with theft detection features," in *Proc. IEEE 14th Int. Conf. Humanoid, Nanotechnol., Inf. Technol., Commun. Control, Environ., Manage. (HNICEM),* 2022, pp. 1–6.

[23] H. Novak, F. Bronić, A. Kolak, and V. Lešić, "Data-driven modeling of urban traffic travel times for short- and long-term forecasting," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 11198–11209, Oct. 2023.

[24] S. D. Nair and S. A. Rahul, "Inverse-impedance-based centralized predictive current controller for n-parallel grid-connected converters for

circulating current elimination," *IEEE Trans. Ind. Electron.*, vol. 70, no. 11, pp. 10848–10859, Nov. 2023.

[25] L. Ge, J. Zhong, Q. Cheng, Z. Fan, S. Song, and R. W. De Doncker, "Model predictive control of switched reluctance machines for suppressing torque and source current ripples under bus voltage fluctuation," *IEEE Trans. Ind. Electron.*, vol. 70, no. 11, pp. 11013–11021, Nov. 2023.

[26] X. Liu, L. Qiu, Y. Fang, K. Wang, Y. Li, and J. Rodriguez, "two-step event-triggered-based data-driven predictive control for power converters," *IEEE Trans. Ind. Electron.*, vol. 71, no. 11, pp. 13545–13555, Nov. 2024.

[27] S. P. Lau, G. V. Merrett, A. S. Weddell, and N. M. White, "A traffic-aware street lighting scheme for smart cities using autonomous networked sensors," *Comput. Elect. Eng.*, vol. 45, pp. 192–207, Jul. 2015.

[28] S. Pizzuti, M. Annunziato, and F. Moretti, "Smart street lighting management," *Energy Efficient*, vol. 6, pp. 607–616, Feb. 2013.

[29] R. Carli, M. Dotoli, and E. Cianci, "An optimization tool for energy efficiency of street lighting systems in smart cities," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 14460–14464, 2017.

[30] L. Xu et al., "Robust predictive current control of hybrid-excited axial flux-switching PM motor based on multiple-resolution parameter identification," *IEEE Trans. Ind. Electron.*, vol. 71, no. 11, pp. 13708–13719, Nov. 2024.

[31] Y. Wei, H. Young, D. Ke, F. Wang, H. Qi, and J. Rodriguez, "Model-free predictive control using sinusoidal generalized universal model for PMSM drives," *IEEE Trans. Ind. Electron.*, vol. 71, no. 11, pp. 13720–13731, Nov. 2024.

[32] Z. Li, J. Xia, X. Gao, Y. Guo, and X. Zhang, "Dual-vector-based predictive torque control for fault-tolerant inverter-fed induction motor drives with adaptive switching instant," *IEEE Trans. Ind. Electron.*, vol. 70, no. 12, pp. 12003–12013, Dec. 2023.

[33] D. Zhou, Z. Zhang, Z. Shen, and J. Zou, "Modulated model predictive control of multisource inverters with flexible power distribution," *IEEE Trans. Ind. Electron.*, vol. 71, no. 11, pp. 13732–13741, Nov. 2024.

[34] Y. He et al., "Direct predictive voltage control for grid-connected permanent magnet synchronous generator system," *IEEE Trans. Ind. Electron.*, vol. 70, no. 11, pp. 10860–10870, Nov. 2023.

[35] K. Zhou, C. Tang, Y. Chen, B. Zhang, and W. Lu, "A generic multi-frequency repetitive control scheme for power converters," *IEEE Trans. Ind. Electron.*, vol. 70, no. 12, pp. 12680–12688, Dec. 2023.

[36] I. Moreno, M. A. No Alejo, T. Saucedo-A, and A. Bugarin, "Modeling led street lighting," *Appl. Opt.*, vol. 53, no. 20, pp. 4420–4430, Jul. 2014.

[37] M. Gapit, H. I. Shaheen, S. Gireesan, and V. Lešić, "Ray tracing model of LED street lighting for energy efficiency simulation," in *Proc. IEEE PES/IAS PowerAfrica*, 2023, pp. 1–5.

[38] T. Novak, K. Pollhammer, H. Zeilinger, and S. Schaat, "Intelligent streetlight management in a smart city," in *Proc. IEEE Emerg. Technol. Factory Automat. (ETFA)*, 2014, pp. 1–8.

[39] V. Lešić, H. Shaheen, M. Gapit, M. Vašak, and H. Kaluder, "A centralized model predictive lighting control method and a centralized prediction-based controllable street lighting system," European Patent Office EP4369866A1, May 2024.

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